Newton's Second Law & Friction: Guided Practice

Name: _____ Period: ____
Date: _____

Equations: $F_{net} = m \cdot a$ | $F_{friction} = \mu \cdot F_N$ | $F_g = m \cdot g$

Constants: $g = 9.8\ m/s^2$ | $1\ N = 1\ kg \cdot m/s^2$

Problem 1

Guided Scaffold

A 4.0 kg cart is pulled across a smooth laboratory track by a horizontal tension cable. Friction between the cart wheels and track is *negligibly small*.

Extracted Givens

- Mass (m) = 4.0 kg
- Horizontal Tension (F_T) = 12.0 N
- Friction force (F_f) = 0 N

[Diagram: Cart on horizontal track pulled to the right by tension F_T with vertical normal force F_N and gravity F_g balanced.]

`F_N (up)` • `F_g = 39.2 N (down)` • `F_T = 12.0 N (right)`

☐ **(a)** **Draw** Construct a labeled free-body diagram for the 4.0 kg cart *while accelerating horizontally*.

FREE-BODY DIAGRAM CANVAS (1 GRID UNIT = 4 N)

• Cart Center (start arrows from center dot)

☐ **(b)** **Calculate** Determine the horizontal acceleration of the cart *assuming zero friction*.

1. SYMBOLIC SETUP	2. SUBSTITUTION & EVALUATION
a = F_net / m	*a = (12.0 N) / (4.0 kg) =* **[m/s^2]**

Problem 2

A student pushes an 8.0 kg wooden crate across a rough linoleum floor with a steady horizontal force of 38.0 N. Kinetic friction opposes the motion.

Extracted Givens

- Mass (m) = 8.0 kg
- Applied Force (F_app) = 38.0 N (right)
- Friction Force (F_f) = 14.0 N (left)

[Diagram: Crate on floor with forward applied push and backward opposing friction.]

`F_app = 38.0 N` • `F_f = 14.0 N` • `m = 8.0 kg`

☐ **(a)** **Calculate** Determine the net horizontal force (F_net) acting on the crate *in the direction of motion*.

1. VECTOR SUM SETUP	2. CALCULATION
	[N]

☐ **(b)** **Calculate** Calculate the resulting horizontal acceleration of the crate *while the push is applied*.

1. SYMBOLIC FORMULA	2. VALUE SUBSTITUTION & ANSWER
	[m/s²]

☐ **(c)** **Explain** Explain what happens to the crate's acceleration if the student *doubles the applied push to 76.0 N* while friction remains 14.0 N.

SCIENTIFIC EXPLANATION (CLAIM & EVIDENCE)